



S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 08

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

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Semester / Year: 4th SEM / 2nd YEAR
Academic Session: 2025-26

Date of Performance:

Date of Submission:

CODE :-

```
M ~  
vansh@vijay MSYS ~  
$ nano note.c  
  
M ~  
GNU nano 8.7  
#include <stdio.h>  
  
int main() {  
    int frames[3];  
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};  
    int time[3];  
    int i, j, k;  
    int hit = 0, fault = 0;  
    int n = 12, frames_count = 3;  
    int counter = 0, flag;  
  
    // initialize frames  
    for(i = 0; i < frames_count; i++) {  
        frames[i] = -1;  
    }  
  
    for(i = 0; i < n; i++) {  
        flag = 0;  
  
        // check page hit  
        for(j = 0; j < frames_count; j++) {  
            if(frames[j] == pages[i]) {  
                hit++;  
                counter++;  
                time[j] = counter;  
                flag = 1;  
                break;  
            }  
        }  
  
        // page fault  
        if(flag == 0) {  
            int min = 0;  
  
            for(j = 1; j < frames_count; j++) {  
                if(time[j] < time[min])  
                    min = j;  
            }  
  
            frames[min] = pages[i];  
            counter++;  
            time[min] = counter;  
            fault++;  
        }  
    }  
  
    float hit_percent = (hit * 100.0) / n;  
    float fault_percent = (fault * 100.0) / n;  
  
    printf("Total Page Hits: %d\n", hit);  
    printf("Total Page Faults: %d\n", fault);  
    printf("Page Hit Percentage: %.2f%%\n", hit_percent);  
    printf("Page Fault Percentage: %.2f%%\n", fault_percent);
```

OUTPUT:-

```
vansh@vijay MSYS ~  
$ ./note  
Total Page Hits: 2  
Total Page Faults: 10  
Page Hit Percentage: 16.67%  
Page Fault Percentage: 83.33%
```